

## Business Analysis Queries

### 1.Total Revenue

```
SELECT SUM(total_amount) AS total_revenue  
FROM orders;
```

|   | total_revenue |
|---|---------------|
| ▶ | 312000.00     |

### 2. Monthly Sales Trend

```
SELECT MONTH(order_date) AS month, SUM(total_amount) AS monthly_revenue  
FROM orders  
GROUP BY MONTH(order_date)  
ORDER BY month;
```

|   | month | monthly_revenue |
|---|-------|-----------------|
| ▶ | 6     | 126000.00       |
|   | 7     | 86000.00        |
|   | 8     | 100000.00       |

### 3. Top Selling Products

```
SELECT p.product_name, SUM(od.quantity) AS total_quantity_sold  
FROM order_details od  
JOIN products p ON od.product_id = p.product_id  
GROUP BY p.product_name  
ORDER BY total_quantity_sold DESC;
```

|   | product_name    | total_quantity_sold |
|---|-----------------|---------------------|
| ▶ | iPhone 14       | 2                   |
|   | Nike Shoes      | 2                   |
|   | Samsung TV      | 1                   |
|   | Laptop          | 1                   |
|   | Washing Machine | 1                   |

### 4.Revenue by Category

```
SELECT p.category, SUM(od.quantity * od.price) AS category_revenue  
FROM order_details od  
JOIN products p ON od.product_id = p.product_id  
GROUP BY p.category  
ORDER BY category_revenue DESC;
```

|   | category        | category_revenue |
|---|-----------------|------------------|
| ▶ | Electronics     | 270000.00        |
|   | Home Appliances | 30000.00         |
|   | Fashion         | 12000.00         |

### 5. Top Customers by Spending

```
SELECT c.customer_id, CONCAT(c.first_name,' ',c.last_name) AS customer_name,
SUM(o.total_amount) AS total_spent
```

```
FROM customers c
```

```
JOIN orders o ON c.customer_id = o.customer_id
```

```
GROUP BY c.customer_id, customer_name
```

```
ORDER BY total_spent DESC;
```

|   | customer_id | customer_name | total_spent |
|---|-------------|---------------|-------------|
| ▶ | 1           | Rahul Sharma  | 156000.00   |
|   | 5           | Vikram Patel  | 70000.00    |
|   | 2           | Priya Mehta   | 50000.00    |
|   | 4           | Sneha Reddy   | 30000.00    |
|   | 3           | Amit Verma    | 6000.00     |

### 6. Customer Lifetime Value (CLV)

```
SELECT c.customer_id, COUNT(o.order_id) AS total_orders, SUM(o.total_amount) AS
lifetime_value
```

```
FROM customers c
```

```
LEFT JOIN orders o ON c.customer_id = o.customer_id
```

```
GROUP BY c.customer_id;
```

|   | customer_id | total_orders | lifetime_value |
|---|-------------|--------------|----------------|
| ▶ | 1           | 2            | 156000.00      |
|   | 2           | 1            | 50000.00       |
|   | 3           | 1            | 6000.00        |
|   | 4           | 1            | 30000.00       |
|   | 5           | 1            | 70000.00       |

### 6. Average Order Value (AOV)

```
SELECT ROUND(AVG(total_amount),2) AS average_order_value
```

FROM orders;

|   |                     |
|---|---------------------|
|   | average_order_value |
| ▶ | 52000.00            |

## 7. Most Profitable Products

```
SELECT p.product_name, SUM((od.price - p.cost) * od.quantity) AS total_profit
```

```
FROM order_details od
```

```
JOIN products p ON od.product_id = p.product_id
```

```
GROUP BY p.product_name
```

```
ORDER BY total_profit DESC;
```

|   | product_name    | total_profit |
|---|-----------------|--------------|
| ▶ | iPhone 14       | 20000.00     |
|   | Samsung TV      | 8000.00      |
|   | Laptop          | 8000.00      |
|   | Nike Shoes      | 5000.00      |
|   | Washing Machine | 5000.00      |